LeetCode Daily Question — October 24, 2026

2409. Count Days Spent Together

Alice and Bob are traveling to Rome for separate business meetings.
You are given four strings: arriveAlice, leaveAlice, arriveBob, and leaveBob.
Alice is in Rome from arriveAlice to leaveAlice inclusive; Bob is in Rome from arriveBob to leaveBob inclusive.
Each date is in the format "MM-DD". Return the total number of days they are in Rome together.

Example 1:
Input: arriveAlice = "08-15", leaveAlice = "08-18", arriveBob = "08-16", leaveBob = "08-19"
Output: 3

Example 2:
Input: arriveAlice = "10-01", leaveAlice = "10-31", arriveBob = "11-01", leaveBob = "12-31"
Output: 0

Approach

Convert each date to its day-of-year number using the non-leap-year month lengths.
Then compute the overlap between Alice's and Bob's inclusive intervals:

start = max(aliceArrive, bobArrive)
end = min(aliceLeave, bobLeave)

If start > end, they do not overlap, so return 0.
Otherwise, return end - start + 1.

Java Solution

```
class Solution {
    private static final int[] DAYS_IN_MONTH = {
        31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31
    };

    public int countDaysTogether(
        String arriveAlice,
        String leaveAlice,
        String arriveBob,
        String leaveBob
    ) {
        int aliceArrive = toDayOfYear(arriveAlice);
        int aliceLeave = toDayOfYear(leaveAlice);
        int bobArrive = toDayOfYear(arriveBob);
        int bobLeave = toDayOfYear(leaveBob);

        int start = Math.max(aliceArrive, bobArrive);
        int end = Math.min(aliceLeave, bobLeave);

        return Math.max(0, end - start + 1);
    }

    private int toDayOfYear(String date) {
        int month = Integer.parseInt(date.substring(0, 2));
        int day = Integer.parseInt(date.substring(3, 5));
```

```
            int dayOfYear = day;

            for (int i = 0; i < month - 1; i++) {
                dayOfYear += DAYS_IN_MONTH[i];
            }

            return dayOfYear;
        }
    }
```

Complexity

Time complexity: O(1). Space complexity: O(1).